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ORIGINAL ARTICLE



Identifying important persons in the promotion of positive behaviour support (PBS) in disability services: A social network analysis

Brent A. Hayward ^a, Lisa McKay-Brown ^a, Shiralee Poed ^b and Keith McVilly ^c

^aMelbourne Graduate School of Education, University of Melbourne, Parkville, VIC, Australia; ^bSchool of Education, University of Queensland, Brisbane, Australia; ^cSchool of Social and Political Sciences, University of Melbourne, Parkville, VIC, Australia

ABSTRACT

Background: Little is known about networks for positive behaviour support (PBS) in disability services. This study explores a PBS network to identify influential persons in its promotion and communication.

Methods: Quantitative sociometric methods were used to identify persons who occupy positions of either boundary spanning (those that link people and groups) and/or opinion leadership (those that are sought for advice).

Results: Nineteen persons were identified. Boundary spanners met all criteria for the position while opinion leaders did not.

Conclusions: This is the first published study of a PBS network in disability services and provides insights into persons involved in its systemic promotion. Boundary spanners are effectively bridging the PBS network, yet persons seeking information about PBS are doing so from opinion leaders that do not display a requisite characteristic for this role. These results have implications for the wider utility of PBS within disability service systems.

KEYWORDS

Boundary spanner; disability; National Disability Insurance Scheme; opinion leader; Positive behaviour support; social network analysis

While positive behaviour support (PBS) offers an informed approach to disability supports, it is hampered by its poor fidelity of implementation and outcomes for people with disability (Hassiotis et al., 2018), leading to a view that it is informally applied in disability services (Bosco et al., 2019). McGill et al. (2018) suggest that to address these practice issues, the focus of PBS intervention should be in the system of supports rather than the individual person with disability, reminiscent of Horner's (2000) assertion two decades prior when he said that the unit of analysis must be expanded to include social systems. Returning to recent years, Horner and Sugai (2018) note that these systems remain largely unaddressed. They recommend investigating the system's variables, which underpin how individuals, groups and organisations make the decision to use and sustain PBS. Rotholz (2009) reflected on what it takes to make PBS successful for people with disability and the systems that are using it. He says that "we cannot rely on the 'the system' or 'the agency' to get this done. We have to be active participants ... working with the organizations responsible for the supports and services" (pp. 130–131). The present Australian study aims to better understand the systems for

PBS through identifying the persons responsible for its promotion.

PBS in Australian disability services

Little research has been conducted to validate the use of PBS in Australian disability services, an observation shared by Beamish and Bryer (2019) of PBS in Australian schools. The literature on PBS in Australian disability services remains limited to outcomes for a handful of people with disability (Gould et al., 2021; Webber et al., 2017), application of training programs (Crates & Spicer, 2012; O'Dwyer et al., 2017), policy analysis (Hayward et al., 2017; 2019a; 2019b; Higgins, 2019), commentary on the relationship between PBS and applied behaviour analysis (Hayward, 2021; Hayward et al., 2018), the use of PBS by occupational therapists (Nott et al., 2019; Perez et al., 2012), developmental educators (Rillotta & Alexander, 2019), and acquired brain injury (ABI) therapists (Carmichael et al., 2019; 2020), the development of a tool (Vassos & Carroll, 2016), the acceptability of PBS in dementia (O'Connor et al., 2019) and ABI (Fisher et al., 2021), and a case study of PBS implementation as part of a State's disability sector (Soo et al., 2017). Despite this

limited literature from Australia, it is noteworthy in that it provides some insight into elements of the PBS system.

A major influence in the promotion of PBS in Australia is the National Disability Insurance Scheme (NDIS; <https://www.ndis.gov.au>). The positive behaviour support capability framework (NDIS Quality and Safeguards Commission, 2019) outlines the limited capabilities required of individuals providing behaviour support in the NDIS: "... the PBS Capability Framework does not outline minimum years of practice, qualifications, training or professional development required to be a behaviour support practitioner" (NDIS Quality and Safeguards Commission, 2019, p. 6). These conceivably low standards allow practitioners with a variety of characteristics, qualifications, experience, and knowledge to use and promote PBS. The framework essentially presents PBS as a method of individual assessment and intervention using behaviour support plans (BSPs). This is not consistent with PBS according to Gore et al. (2013), who specify that BSPs constitute only one of ten key components of PBS.

There is emerging evidence that service providers in the NDIS, which includes behaviour support practitioners, are poorly networked, risking disruption of the disability sector and poor support quality (Malbon et al., 2017). So, following recent recommendations in the literature to address the systems in which PBS is used by individuals, groups, and organisations, and building upon the Australian research to-date, this study explores the network of PBS in one Australian State to identify influential persons in its communication and promotion.

Social network analysis

A successful way of exploring the relationships between people, groups, and organisations in a system is social network analysis (SNA). SNA uses special software and mathematical algorithms to describe the pattern of relationships within a system. Importantly, SNA provides a method for analysing formal and informal information flows in systems, helping to understand the nature of relations between components in a system, and how these relationships influence its structure (Blanchet & James, 2012).

The insights provided by the Australian PBS research to-date suggests a variety of persons, from different organisations, working with different groups of people with disability, providing a variety of services in the context of the NDIS constitutes the network for PBS in Australian disability services. No studies to date have

explored networks for PBS in a disability context, and no studies have explored PBS at a state network level or larger. In order to understand the PBS network effectively, we need to know which persons are integral to its functioning. As opinion leaders and boundary spanners are key persons within professional social networks (Cranley et al., 2019), we use these constructs to examine the communication of PBS.

Opinion leaders and boundary spanners

Opinion leaders are the persons that are asked for advice and can influence others. They are typically local, influential, credible, likeable, trustworthy, and educationally influential. But their role is not a function of their status. Trepte and Scherer (2010) argue that opinion leaders consciously select and transmit information to others. This can have implications for the type and quality of information they are communicating and because of this, opinion leaders have been the focus of social network studies in a variety of fields.

The second type of person are boundary spanners, also referred to as bridgers and brokers (Long et al., 2013). Boundary spanners link two other persons or groups that are not themselves linked. They control the information flow between groups of people in the network to a higher extent than others (Bodin et al., 2006). Boundary spanners can facilitate access to new information, transfer knowledge, and coordinate efforts across the network (Long et al., 2013). They are particularly useful in client-centred models of service planning and delivery because of the interactions between different agencies and professions (Williams, 2011), thereby making them valuable considerations in studies of disability services.

Opinion leaders and boundary spanners are mathematically identified in social network analyses. Opinion leaders are commonly identified using the measure of *indegree centrality*, which identifies those with the highest number of incoming ties from others in the network. Boundary spanners are commonly identified using the measure *betweenness centrality*, which is the extent to which an actor lies between other actors that would not otherwise be connected (Cranley et al., 2019). These two roles are collectively referred to as *network positions*.

Identifying the population to be studied

The disability workforce in the NDIS is described as heterogeneous (Vassos et al., 2019). Behaviour support in the context of PBS in this NDIS mirrors this, with behaviour support plans written by practitioners and

implemented by support workers (Vassos & Carroll, 2016). As a result, this population can be invisible and hard-to-reach. Such populations are sometimes called “hidden populations” in social network analysis and can pose a challenge to study.

In order to establish the boundary for sampling, network analysis constructs the social boundaries of the population rather than sampling it. The present study uses a relational approach to specify the boundary of the population using a reputational snowballing method (addressed separately in this paper). The inclusion of persons in the population is specified in terms of a person’s presence in the network, their attributes, and the content of their relationships (Laumann et al., 1983). This method produces a complete network of all the relationships among a set of respondents in a specified boundary. In the present study it is those persons who are communicating about PBS in the state of Victoria, Australia. In this way, the PBS network is constructed via social boundaries.

Snowball sampling is used in the present study. It is a type of non-probability sampling used when eligible participants are difficult to identify and locate (Williamson, 2018), i.e., “hidden”. Snowball sampling permits identification of participants when the boundaries of the study are socially defined through asking them to self-identify their involvement in the phenomenon of interest. The network we are interested in is therefore considered unbounded in social network analysis terms. As a non-probability sample, snowballing does not produce generalisable results. Instead, snowballing continues until a deep, rich data set is obtained (Williamson, 2018).

Study aims

- (1) Identify the boundary spanners and opinion leaders in the PBS network.
- (2) Evaluate if the characteristics of these persons are consistent with these research-validated network positions.
- (3) Describe how the results may impact the utility of PBS in the NDIS.

Methods

As there is no endorsed tool for assessing the quality of studies using social network analyses, we applied the guidance of Sabot et al. (2017), Appendix 1, and the reporting guidelines for survey research described by Bennett et al. (2011), Appendix 2.

Ethics approval

Ethical approval for this study was granted by the Melbourne Graduate School of Education Human Ethics Advisory Group (1852429).

Context

The results reported here are part of a larger study conducted over three months in 2020 in the state of Victoria, Australia. Data were collected using an online survey on the Qualtrics® platform. A bespoke website was established, and the landing page welcomed invitees, provided the contact details of the researchers, presented the plain language statement, and recorded participant consent. A validation procedure using assigned participant identification numbers ensured that only those persons invited to participate could do so, and only once. After consent was provided, participants were then taken to the survey questions in Qualtrics. Upon completion of the survey, participants were automatically directed back to the website where they could opt-in to a draw for an incentive.

Sampling procedure

Seed sample

An initial group of participants who believed they know, and could name, persons involved in the use and promotion of PBS constituted the first set of participants, commonly referred to as the seed sample. The seed sample was established using staff from the state government office with responsibility for oversight of disability services. The legislative function of this office and its staff provides confidence that they would have knowledge of PBS and know which persons in Victorian disability services may be using and promoting it. Using a recall name generation method and similar to Buchthal et al. (2013), seed sample participants were asked to list the names and the employing organisations only of persons they *believe* to be using and promoting PBS in Victorian disability services. Participants were not required to specifically know this; it was their perception only. No clarification of the term “PBS” was provided, in order to allow participants to provide their own interpretation, consistent with the method of Cullerton et al. (2017) and consistent with the NDIS, which does not consistently define PBS (Hayward et al., 2019). The lists of individual names were pooled to form a final list of names.

Snowball sample

Persons in the final list developed from the seed sample became those participants in the first wave of the snowball. Because participants were not asked to provide any

contact details for their nominees, each named nominee was searched online to locate their email address, or their email address was discovered from online sources. If a nominee's email address could not be located online or ascertained from online sources, then they were excluded from the first wave. Seeking email addresses for nominees directly from respondents would have created a complex ethical scenario requiring seeking consent from nominees, which would have undermined the research methodology and produced distorted results (see Borgatti & Molina, 2003).

Each wave invitee was allocated a unique identification number to replace their name during data collection and analysis. This identification number was specified in an email from the first author, inviting them to participate in the survey, and directing them to the website. Invitees were sent one reminder email after two weeks if they did not complete the survey. If they did not complete the survey after another two-week period, then they were removed from the study. When responses to the snowball "dried out", the wave was arbitrarily deemed over and the names of respondents and nominees were manually cleaned and a new list of survey invitees was constructed (Contandriopoulos et al., 2019). The waves of snowball sampling continued in this manner (Figure 1). Persons who were nominated but who did not respond to the survey were retained for the purpose of graphing the network, as excluding these persons would produce misleading results (Klov Dahl, 2005).

Survey

Survey structure

Sociodemographic questions (Appendix 3) were informed by Glegg et al.'s (2019) scoping review of networks and knowledge translation and Tasselli's (2014) review of social networks in healthcare organisations. The survey was piloted by three persons familiar with PBS but not considered potential respondents to the survey, and minor modifications were made. The survey included a total of 48 questions of which only 19 are addressed in this paper. Using a reputational method, equivalent to a name generator in social network analysis (Lewis, 2006), participants in each wave were asked to enter the names of up to five persons from whom *they sought* advice or information about PBS and the organisation for which each person worked. Participants were also asked the same questions for people who sought advice or information *from them* about PBS however only the data relating to the advice and information *they sought* is presented here.

Data extraction and analysis

Survey responses were exported from Qualtrics[®] into a spreadsheet where the data were cleaned following a two-step process. First, participants' identification numbers were checked against a master list to ensure that only the data from invited respondents were included. In situations where a participant's data were recorded twice, only the first record was retained. Second, only participants who completed the sociodemographic survey questions and listed at least one person using their full name as a source of advice or information about PBS were included.

Social network data in the form of adjacency lists were created and imported into the social network software Visone (Algorithmics Group, 2019) where graphs were constructed and analyses conducted, including calculation of indegree and betweenness centralities.

Identification of important positions in the network

Wasserman and Faust (1994) suggest using the mean to summarise the centralities of all the persons in the network. Once the means of the indegree and betweenness centralities have been calculated, Himelboim et al. (2017) recommend identifying the single point in the distribution of each centrality score where there is a natural drop in the mean from high to low. The score above this point then becomes the cut-off score for boundary spanners and opinion leaders.

Criteria for important network positions

The ways in which boundary spanners are identified vary greatly (Long et al., 2013). Criteria for boundary spanners were developed from Cranley et al. (2019) and Jesiek et al. (2018). One criterion could not be explored using self-reported data (discipline/domain expertise) and was excluded. The knowledge criterion is addressed later in this paper. Criteria for opinion leaders were taken from Rogers (2003), however the criterion for innovativeness could not be explored using demographic data and was excluded; Table 1. Persons with professional university degrees were established from their job title, with clinical, education, and academic roles considered to meet this criterion. Social status was taken from job type and considered to be those persons who were reported to be in government and university roles. Outdegree centrality was calculated in Visone (Algorithmics Group, 2019). For the purposes of this study, outdegree centrality is described as the number of links out from a person in the network. Outside orientation for each respondent in the network was confirmed by the report of at least one network link

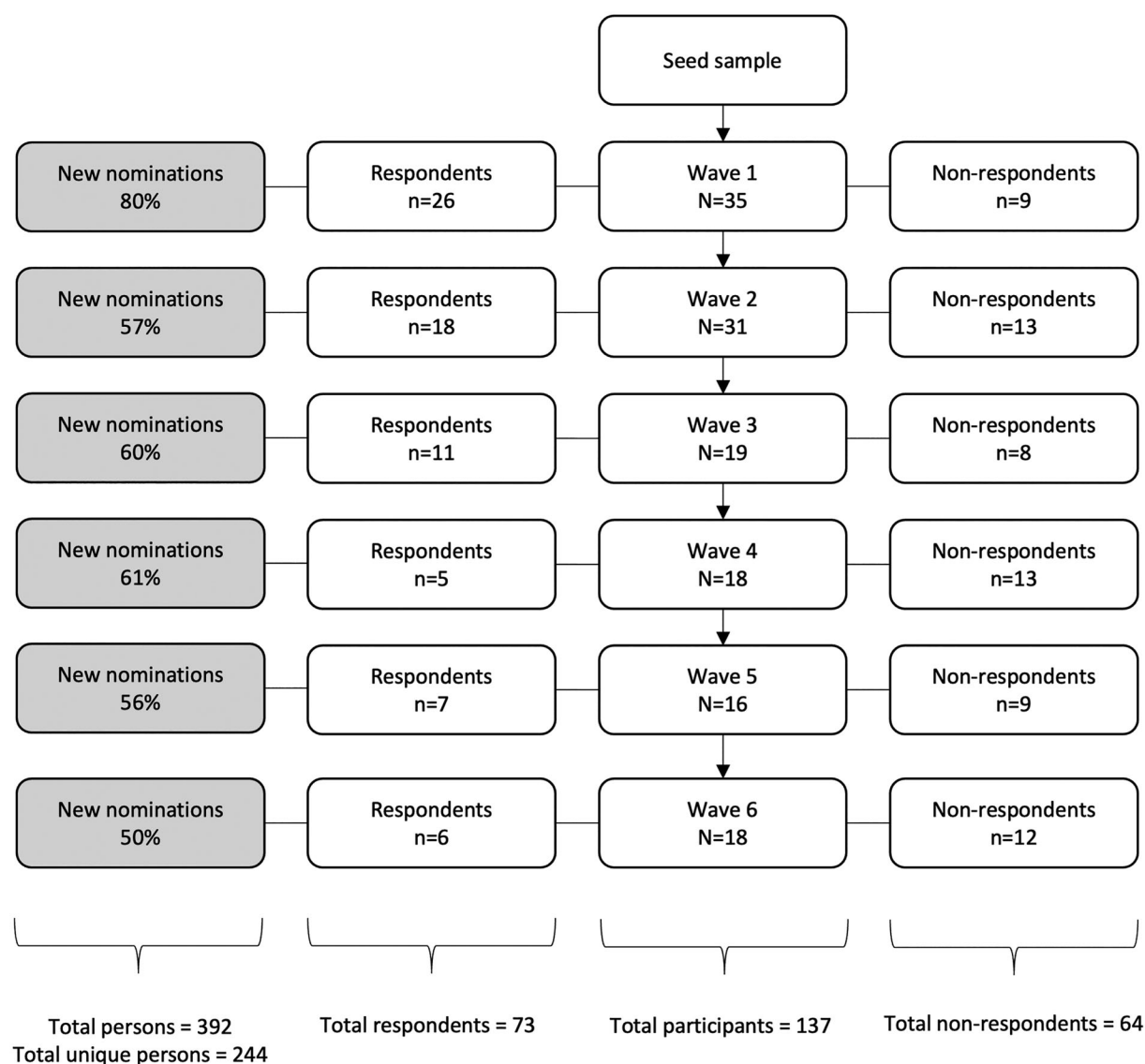


Figure 1. Snowball waves with number of respondents, non-respondents, and proportion of new nominations.

Note: Data collection ceased at wave 6 when all persons were located outside the boundary of the study.

outside their own organisation; Table 2. Years working in current role and in disability sector was calculated by identifying those indicating ≥ 10 years of tenure. Weighted scores were used to compare the reported sources of information by converting each response to a numeral using weights: daily (4), weekly (3), monthly (2), less than monthly (1), and never (0), Appendix 4.

The criteria met for each identified boundary spanner and opinion leader was tallied (Table 3).

Results

Six snowball waves were conducted (consistent with Lewis, 2006), ceasing when all respondents were located outside the region under study, consistent with Freitag

et al. (2018); Figure 1. Forty-four unique persons were included in the seed sample. A total of 136 persons were invited to participate across six waves of snowballing. A total of 73 persons responded to the survey (response rate = 54%), with 42 (58%) of those persons participating upon the initial invitation. The survey took an average of 22 min to complete. The number of invitees for each wave ranged between 16 and 35 persons (mean = 23). Up to 26 persons responded at each wave (mean = 12). Nominations of new persons varied between 50 and 80%. Of those who responded but were excluded from the analysis, five had insufficient details to identify persons nominated, two did not list any person, and one was duplicated. A total of 66 participants had sufficient data to include in the analysis.

Table 1. Criteria for boundary spanners and opinion leaders with the associated data source for this study.

Network position	Criteria	Data source
Boundary spanner	Lifelong learners	Sources of information
	Long tenure in the field	Years in the disability sector
	Long tenure in current job	Years in current role
	Technical expertise	Job type
	Greater discipline/domain experience ^a	-
	Greater knowledge ^b	-
Opinion leader	Greater exposure to mass media	Sources of information
	Greater contact with change agents	Ties with actors whom have a technical university degree, with social status
	Greater social participation	Outdegree (network measure)
	More cosmopolite	Orientation outside one's network
	Higher socioeconomic status	Job type
	More innovative ^a	-

Note: Criteria developed from Cranley et al., 2019; Jesiek et al., 2018; Rogers, 2003.

^aCould not be explored in this study and was excluded from the analysis. ^bThis criterion will be examined in a subsequent study, see conclusion.

Network graph

The network graph (Figure 2) consists of a main graph and two subgraphs, the latter in the top right-hand and bottom left-hand corners. The top sub-graph represents all respondents working in the acquired brain injury sector while the bottom is those respondents from a single disability service provider organisation. The main graph resembles a global snowflake structure with nine points (see Feng & Porter, 2020).

Indegree and betweenness centrality

The median indegree score was 0.52 and the mean was 0.71. Both the median and the mean occur at a single point in the distribution of all indegree centrality scores between 0.52 and 1.04. A natural drop from high to low in the distribution of indegree centrality scores was present at this point, therefore all persons with indegree centrality scores >1.04 are considered to be opinion leaders. This is equivalent to 11

persons in the network and these persons are identified as triangles in the network graph (Figure 2) indicating their role as opinion leaders in the network. Six persons identified as having high indegree centrality and who did not respond to the survey are described in Table 4.

The median betweenness centrality score was 1.27 and the mean was 0.71. The mean and median occur at a single point in the distribution of all betweenness centrality scores with a drop from high to low between 0.59 and 1.33, therefore all persons with betweenness centrality scores > 1.33 are considered boundary spanners. This is equivalent to nine persons in the network, all of whom were respondents to the survey. These persons are identified as squares in the network graph indicating their role as boundary spanners in the network. Six persons (all respondents to the survey) occupy both opinion leader and boundary spanner roles and are represented as octagons in the network graph (Figure 2).

Table 2. Additional characteristics used in the identification of network positions.

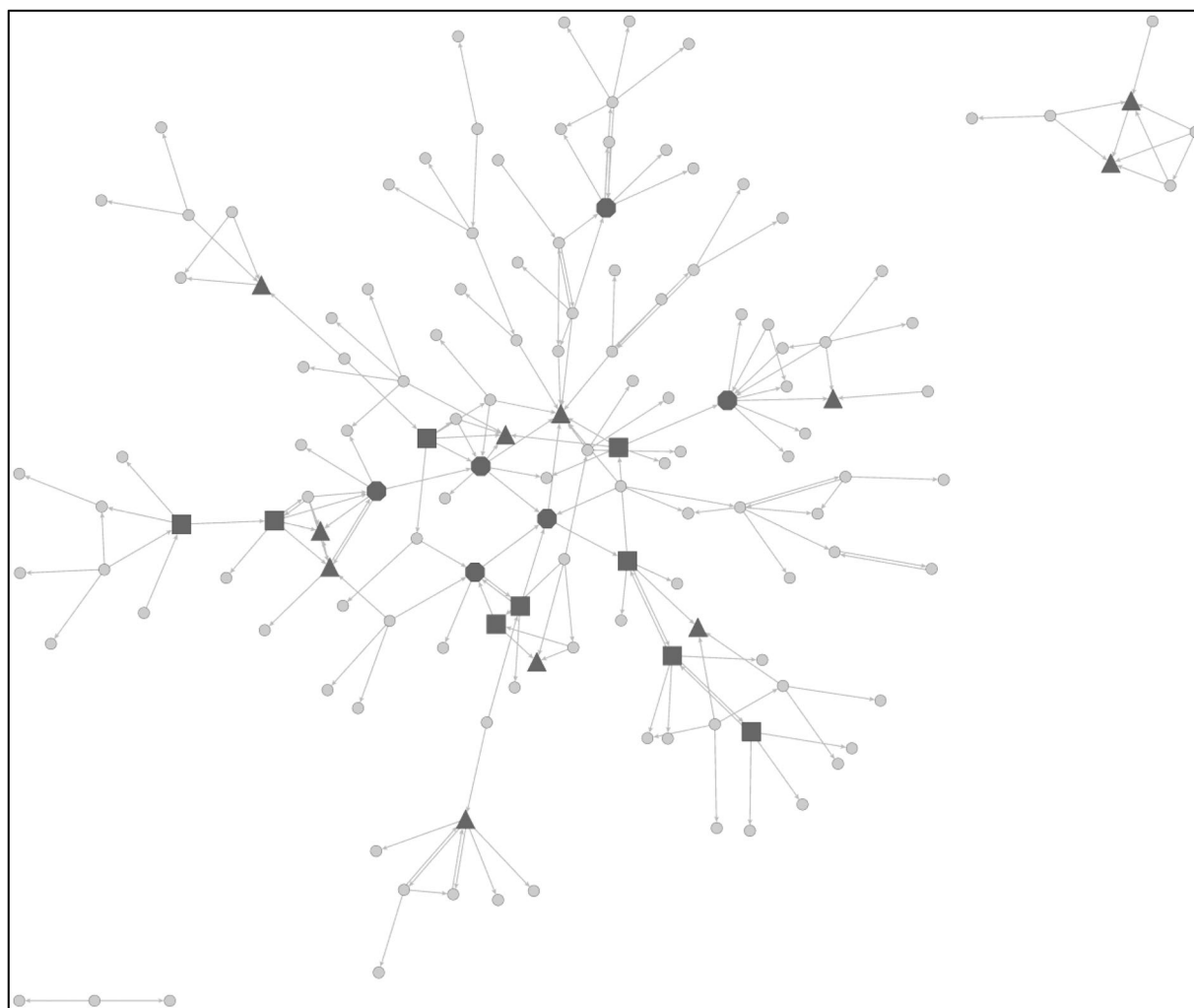
Respondent	Indegree centrality	Betweenness centrality	Outdegree centrality	External orientation	Tie status
48	2.083	1.333	2.604	Y	T, U
40	-	1.373	2.604	N	T, G
30	-	1.373	1.563	Y	T, G
06	-	2.157	1.042	Y	G
53	-	2.824	2.083	Y	U
09	-	2.902	1.563	Y	U, I
36	-	4.039	2.604	Y	-
05	2.083	5.490	1.563	Y	I, T
41	2.083	5.608	2.604	Y	T, U
66	1.563	6.824	2.604	Y	G, T
28	-	7.020	2.604	Y	U, T
04	-	7.922	2.604	Y	U, I
17	2.083	10.706	2.604	Y	U, T
26	2.083	17.373	1.042	Y	U, T
21	-	18.196	2.604	Y	G, I, T
01	1.563	-	0.521	Y	T
43	1.563	-	0.521	N	-
11	2.083	-	1.563	Y	I, T
46	2.083	-	0.521	Y	I, T

Note: Y = yes; N = no; T = technical; U = university appointment; I = internationally known; G = government role; - = none.

Table 3. Criteria summary of individual opinion leaders, boundary spanners and combined positions.

Respondent	Role	Characteristic						
		Combined		Opinion leader			Boundary spanner	
		Job type	Sources of information	Outdegree high	External orientation	Ties with technicians	Tenure in current role	Tenure in disability sector
40	Boundary spanner	Y	Y	-	-	-	N	N
30	Boundary spanner	N	N	-	-	-	Y	Y
06	Boundary spanner	N	N	-	-	-	N	Y
53	Boundary spanner	Y	N	-	-	-	N	Y
09	Boundary spanner	N	Y	-	-	-	Y	N
36	Boundary spanner	N	Y	-	-	-	N	Y
28	Boundary spanner	Y	Y	-	-	-	N	N
04	Boundary spanner	Y	N	-	-	-	N	Y
21	Boundary spanner	Y	Y	-	-	-	N	Y
01	Opinion leader	Y	Y	N	Y	Y	-	-
43	Opinion leader	Y	Y	N	N	N	-	-
11	Opinion leader	Y	N	N	Y	Y	-	-
46	Opinion leader	Y	N	N	Y	Y	-	-
17	Combined	Y	N	Y	Y	Y	N	Y
26	Combined	Y	N	N	Y	Y	N	Y
48	Combined	N	Y	Y	Y	Y	N	Y
05	Combined	Y	N	N	Y	Y	N	Y
41	Combined	Y	Y	Y	Y	Y	Y	Y
66	Combined	Y	Y	Y	Y	Y	Y	Y

Note: Y = Yes; N = No; - = not applicable.

**Figure 2.** The PBS network graph.

Note: Dark nodes are important network positions: triangles are opinion leaders, squares are boundary spanners, and octagons are combined opinion leader and boundary spanners. Subgraphs are in the top right-hand and bottom left-hand corners.

Table 4. Characteristics of non-respondents with opinion leadership positions in the network.

Non-respondent	Indegree centrality	Job type
A	1.56	Government
B	1.56	Support organisation
C	1.56	Private
D	2.08	University
E	2.08	Private
F	5.21	University

Outdegree centrality

The mean outdegree centrality score for opinion leaders was 0.782, while that for persons identified as both opinion leaders and boundary spanners was 2.170. Opinion leaders had the same number of ties with actors from universities but less ties with persons from government compared with boundary spanners.

Description of those persons with important network positions

Table 3 summarises the results of the network position criteria reported by the 19 persons in identified network positions, i.e., boundary spanners, opinion leaders, and combined. As separate groups, criteria for boundary spanners and combined roles were present, while opinion leaders failed to meet one criterion (Table 5).

Boundary spanners

Boundary spanners had primarily clinical roles but were also the only persons who identified as having educational roles, were the only persons from government and support organisations, and were the only group to include persons with PhD or doctorates. Along with persons with combined roles, boundary spanners had more professional memberships. They had a more varied geographical scope of work, worked from a variety of locations, and were more likely to not work directly with people with disabilities. Boundary spanners as a group are characterised by their job type, sources of information, memberships, education, and years working in the disability sector. Whilst all criteria were present across the boundary spanners, the *tenure in current role* criterion was the lowest recorded.

Opinion leaders

Opinion leaders reported primarily “other” roles and reported no education or management roles. They commonly worked in private practice, reported no professional memberships, worked at a state level, and were the only group to report working from a regional location, and overall work from within Victoria. They had a wide range of work responsibilities. All reported working directly with adult persons with intellectual disability and were the only group to report working with people with ABI. As a group, opinion leaders are characterised by their job type, external orientation, and ties with technicians, however opinion leaders failed to meet the *high outdegree* criterion.

Combined boundary spanners and opinion leaders

This group reported no educational or academic roles nor worked in support organisations or universities. They typically had a greater number of professional memberships, worked across multiple states, had worked in their current role for a greater range of years, and worked from a greater variety of locations. The majority worked with adult persons with intellectual disability or did not work directly with people with disability. In this study, being a combined opinion leader and boundary spanner contributed to higher outdegree, external orientation, ties with technicians, and tenure in the disability sector.

Discussion

PBS boundary spanners

Boundary spanners had variable characteristics, which support their linkage role in the PBS network. This

Table 5. Summary of results of network position criteria.

Criteria	Combined	Opinion leaders	Boundary spanners
Job type	Y	Y	Y
Sources of information	Y	Y	Y
High outdegree centrality	Y	N	-
External orientation	Y	Y	-
Ties with technicians	Y	Y	-
Tenure in current job	Y	-	Y
Tenure in disability sector	Y	-	Y

Note: Y = Yes; N = No; - = not applicable.

was exemplified by the three boundary spanners with the highest betweenness centrality (respondents 17, 21, and 26) whose betweenness centrality scores were also at least two standard deviations above the mean betweenness centrality score. These three persons shared few common characteristics; they worked in the disability sector for different periods of time, worked across different jurisdictions, and worked from different locations. They were networked across an entire spectrum of geographies, from Victoria, through other states, and internationally and operated within government, university, and the private sectors. This variation is effective for boundary spanners to link heterogeneous persons and groups in the PBS network (Hayward et al., 2018) that are not themselves linked, helping to facilitate access to new information, transfer knowledge, and coordinate efforts across the network.

The network analysis did not identify any non-respondents as boundary spanners, suggesting that snowball sampling effectively reached boundary spanners in this network. This was not unexpected as boundary spanners occupy coordinating positions in networks, foreseeably facilitating snowball sampling.

As a group, boundary spanners in this PBS network met all the criteria for the position however *long tenure in current job* (greater than 10 years) was recorded by the least number of boundary spanners. This result may be related to the immaturity of the NDIS service system and its promotion of PBS, thereby not permitting sufficient time for boundary spanners to be established in their job and to then meet this criterion more consistently.

PBS opinion leaders

Opinion leaders were more homogenous. They had less descriptive professional titles and typically worked privately. This was not surprising as the NDIS is now awash with private practitioners. Interestingly, a majority of opinion leaders had no professional memberships, suggesting they are less connected to others and to contemporary information. This is supported by Adams and Barron (2009) who found that professional membership was a significant contributor to the use of evidence-based practices by school nurses. Opinion leaders typically worked at a state level and were the only group to include persons working from regional locations, consistent with the devolution of responsibilities in the NDIS from the Federal to the State and Territory governments, thereby positioning these persons as sources of information about PBS. All opinion leaders worked exclusively with adults with disability. Opinion leaders had a variety of work

responsibilities including education, providing information and advice, assessment and planning, and research, representing a broad set of skills generally considered important for PBS opinion leaders.

The network analysis identified six additional opinion leaders who were not respondents to the survey (Table 4). One of these persons had the highest indegree centrality in the entire network (5.21) and from the data specified about this person from the survey, they worked in a university. However, none of the respondent opinion leaders in this PBS network met the *high outdegree* criterion for this position. We argue that this is a result of the low standards required for a person to be considered a behaviour practitioner in the NDIS. It is important to also note that these standards are related to behaviour assessment and BSP writing, not a comprehensive application of PBS. Fewer opinion leaders met all the specific criteria for this role (i.e., high outdegree, external orientation, ties with technicians) compared to persons in combined roles. This suggests that opinion leadership functions in this network are supplemented by boundary-spanning functions.

Opinion leadership stems from two sources: prominent knowledge (“expert opinion leader”) and personality traits (“social opinion leader”), that operate in a complementary fashion (Gnambs & Batinic, 2013). It is unclear from this study which type of opinion leadership may be more prominent in this PBS network as we have yet to evaluate knowledge. However, it seems more plausible that this network functions with expert opinion leaders because of two things. First is the lack of high outdegree centrality, which indicates low social connectedness. The second is the social status inferred by the “behaviour support practitioner” title in the NDIS. This latter point is supported by the demographic characteristics of opinion leaders who commonly have “other” job titles and work privately, all indicative of behaviour support practitioners in the NDIS. Despite the social status that may be inferred, staff in disability support services (e.g., Bradshaw & Goldbart, 2013) and parents (e.g., McGill et al., 2006) have expressed concerns about the usefulness of such “expert” services. This underscores the importance of accurate knowledge held by opinion leaders in this PBS network.

Strengths and limitations

The present study yielded a response rate of 55%. This is a fair result in comparison to studies with similar methodologies (e.g., Farquharson, 2005; Kowald & Axhausen, 2012; Lewis, 2006; Parsons, 2018), some of which used easily defined seed samples, used organisations rather than individual informants, distributed surveys via senior

persons, used interviews to gather network data, and some with extensive follow-up procedures. Of worth noting is Malbon et al.'s (2017) study of service providers in the NDIS in two defined geographical areas of Australia. While that study did not use snowball methodology, it did use social network analysis methods to explore the information the providers access through their networks. Unfortunately, neither the number of organisations sampled, nor the response rates are reported, however the sample sizes were described as “reasonable” ($n = 29$) and “low” ($n = 9$). The sample size in the present study ($n = 65$) is higher than that in Malbon et al.'s study. While Kowald and Axhausen (2012) outline sources of selection bias in snowball sampling, SNA overcomes these as a random sample is not desired.

The exclusion of persons for whom an email address could not be identified is a bias that could not be overcome within the ethical constraints of this study as described in the methods section. The manual identification of invitees' email addresses (including those which were not available publicly) and matching referred persons through the survey may be vulnerable to human error however several checks were implemented to minimise this risk. Establishing criteria for boundary spanners is problematic because of the variation in the ways this position is identified in the literature however we sought criteria from two sources; one of a similar service system (long-term care; Cranley et al., 2019) and a review of a professional field (Jesiek et al., 2018).

This study used quantitative sociometric methods to identify persons in important network positions and self-reported survey data to quantify criteria associated with these positions. We acknowledge that only a small number of opinion leaders were included in the analysis. It may be the case that additional opinion leaders are present in the network; however, this study did not identify them because they were not nominated by respondents. Indegree and betweenness centralities should be interpreted with caution as results may not be representative of the complete network, and this network is a snapshot in time and may not be representative of the network in the future nor a PBS network elsewhere, therefore the results are not generalisable. Finally, it is important to consider that respondents and their nominees may in fact be communicating about BSPs and not PBS, reflecting the limited way in which PBS is described by the NDIS explained in the introduction.

Conclusion

This study has helped to explore some of the PBS system variables by identifying the persons responsible for its promotion. PBS boundary spanners were well-

articulated in this network, and were effectively bridging the network, with the exception of the brain injury PBS sector. As a group, they met all criteria for this position. Some individual boundary spanners met more criteria than others, and we speculate this to be related to the infancy of the NDIS, which has not permitted this PBS network to develop sufficiently. Conversely, PBS opinion leaders as a group did not meet all criteria for this position. In effect, persons seeking information about PBS were doing so from people that do not display all the requisite characteristics for this role and as an extension, may not necessarily have expertise in PBS. For example, PBS opinion leaders did not access more sources of information about PBS, were not linked to more persons overall, nor to more persons outside their own organisation.

Johnson (2012) is careful to highlight that opinion leaders may exhibit poor fidelity for complex interventions for which they have only limited understanding. We suggest this is an important consideration because fidelity is necessary for the implementation of PBS (versus only BSP development), thereby potentially leading to variation in the use and promotion of PBS by opinion leaders. This is likely influenced by the comparatively low standards required for a person to be considered a behaviour practitioner in the NDIS. We hypothesise that the opinion leaders in this PBS network are more consistent with the “expert opinion leader” type described by Gnambs and Batinic (2013) because of the status conferred by their role in the network. Further research should consider replicating this study in another state or territory to examine if these findings are more widespread. We suspect the development of a functional PBS network in Victoria as part of the NDIS will take additional time. How long that is depends on when and if the NDIS behaviour support practitioner workforce matures and stabilises.

We acknowledge that these results do not take into account the perspectives of individuals identified as boundary spanners and opinion leaders. Therefore, we are currently analysing additional data collected as part of this study to investigate if boundary spanners and opinion leaders report knowledge, attitudes and leadership characteristics consistent with these positions. This will assist to explore the remaining boundary spanning criteria, which could not be addressed in this study, and more confidently describe the type of opinion leaders in this network.

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ORCID

Brent A. Hayward  <http://orcid.org/0000-0003-1189-2756>

Lisa McKay-Brown  <http://orcid.org/0000-0001-6269-7278>

Shiralee Poed  <http://orcid.org/0000-0001-6038-2184>

Keith McVilly  <http://orcid.org/0000-0002-9753-9031>

References

- Adams, S., & Barron, S. (2009). Use of evidence-based practice in school nursing: Prevalence, associated variables, and perceived needs. *World Views on Evidence-Based Nursing*, 6(1), 16–26. <https://doi.org/10.1111/j.1741-6787.2008.00141.x>
- Algorithmics Group. (2019). *Visone* (Version 2.18) [computer software]. University of Konstanz. <http://visone.ethz.ch/>
- Beamish, W., & Bryer, F. (2019). Behavioural support in Australia. In F. Bryer & W. Beamish (Eds.), *Behavioural support for students with special education needs. Advancing inclusive and special education in the Asia-Pacific* (pp. 69–88). Springer. <https://doi.org/10.1007/978-981-13-7177-6>
- Bennett, C., Khangura, S., Brehaut, J. C., Graham, I. D., Moher, D., Potter, B. K., & Grimshaw, J. M. (2011). Reporting guidelines for survey research: An analysis of published guidance and reporting practices. *PLoS Medicine*, 8(8), e1001069. <https://doi.org/10.1371/journal.pmed.1001069>
- Blanchet, K., & James, P. (2012). How to do (or not to do) ... a social network analysis in health systems research. *Health Policy and Planning*, 27(5), 438–446. <https://doi.org/10.1093/heapol/czr055>
- Bodin, O., Crona, B., & Ernstson, H. (2006). Social networks in natural resource management: What is there to learn from a structural perspective? *Ecology and Society*, 11(2), r2. <https://doi.org/10.5751/ES-01808-1102r2>
- Borgatti, S. P., & Molina, J. L. (2003). Ethical and strategic issues in organizational social network analysis. *Journal of Applied Behavioral Science*, 39(3), 337–349. <https://doi.org/10.1177/0021886303258111>
- Bosco, A., Paulauskaite, L., Hall, I., Crabtree, J., Soni, S., Biswas, A., Cooper, V., Poppe, M., King, M., Strydom, A., Crawford, M. J., & Hassiotis, A. (2019). Process evaluation of a randomised controlled trial of PBS-based staff training for challenging behaviour in adults with intellectual disability. *PLoS ONE*, 14(8), e0221507. <https://doi.org/10.1371/journal.pone.0221507>
- Bradshaw, J., & Goldbart, J. (2013). Staff views of the importance of relationships for knowledge development: Is training by specialists a waste of money? *Journal of Applied Research in Intellectual Disabilities*, 26(4), 284–298. <https://doi.org/10.1111/jar.12020>
- Buchthal, O. V., Taniguchi, N., Iskander, L., & Maddock, J. (2013). Assessing state-level active living promotion using network analysis. *Journal of Physical Activity and Health*, 10(1), 19–32. <https://doi.org/10.1123/jpah.10.1.19>
- Carmichael, J. J., Gould, K. R., Hicks, A. J., Feeney, T. J., & Ponsford, J. L. (2019). Understanding Australian community ABI therapists' preferences for training in and implementing behaviour interventions: A focus on positive behaviour support. *Brain Impairment*, 21(2), 191–207. <https://doi.org/10.1017/BrImp.2019.32>
- Carmichael, J. J., Hicks, A. J., Gould, K. R., Feeney, T. J., Analytis, P., & Ponsford, J. L. (2020). 'We struggle and muddle.' A qualitative study exploring community ABI therapists' experiences of using, training in and implementing behaviour interventions. *Brain Impairment*, 22(1), 1–16. <https://doi.org/10.1017/brimp.2020.6>
- Contandriopoulos, D., Sapeha, H., & Larouche, C. (2019). Some insights related to social network analysis data collection challenges – a research note. *International Journal of Social Research Methodology*, 22(5), 463–468. <https://doi.org/10.1080/13645579.2019.1574957>
- Crangle, L. A., Keefe, M., Taylor, D., Thompson, G., Beacom, A. M., Squires, J. E., Estabrooks, C. A., Dearing, J. W., Norton, P. G., & Berta, W. B. (2019). Understanding professional advice networks in long-term care: An outside-inside view of best practice pathways for diffusion. *Implementation Science*, 14(1), 10. <https://doi.org/10.1186/s13012-019-0858-6>
- Crates, N., & Spicer, M. (2012). Developing behavioural training services to meet defined standards within an Australian statewide disability service system and the associated client outcomes. *Journal of Intellectual and Developmental Disability*, 37(3), 196–208. <https://doi.org/10.3109/13668250.2012.703318>
- Cullerton, K., Donnet, T., Lee, A., & Gallegos, D. (2017). Joining the dots: The role of brokers in nutrition policy in Australia. *BMC Public Health*, 17(1), 307. <https://doi.org/10.1186/s12889-017-4217-8>
- Farquharson, K. (2005). A different kind of snowball: Identifying key policymakers. *International Journal of Social Research Methodology*, 8(4), 345–353. <https://doi.org/10.1080/1364557042000203116>
- Feng, M., & Porter, M. A. (2020). Spatial applications of topological data analysis: Cities, snowflakes, random structures, and spiders spinning under the influence. *Physical Review Research*, 2(3), 033426. <https://doi.org/10.1103/PhysRevResearch.2.033426>
- Fisher, A., Bellon, M., Lawn, S., & Lennon, S. (2021). Family perspectives on the acceptability and usefulness of the FAB positive behaviour support program: A pilot study. *Brain Injury*, 35(5), 609–619. <https://doi.org/10.1080/02699052.2021.1894479>
- Freitag, A., Vogt, B., & Hartley, T. (2018). Breaking stereotypes through network analysis of the Chesapeake oyster community. *Marine Policy*, 90, 146–151. <https://doi.org/10.1016/j.marpol.2017.12.023>
- Glegg, S. M., Jenkins, E., & Kothari, A. (2019). How the study of networks informs knowledge translation and implementation: A scoping review. *Implementation Science*, 14, 34. <https://doi.org/10.1186/s13012-019-0879-1>
- Gnambs, T., & Batinic, B. (2013). The roots of interpersonal influence: A mediated moderation model for knowledge

- and traits as predictors of opinion leadership. *Applied Psychology*, 62(4), 597–618. <https://doi.org/10.1111/j.1464-0597.2012.00497.x>
- Gore, N. J., McGill, P., Toogood, S., Allen, D., Hughes, J. C., Baker, P., Hastings, R. P., Noone, S. J., & Denne, L. D. (2013). Definition and scope for positive behavioural support. *International Journal of Positive Behavioural Support*, 3(2), 14–23.
- Gould, K. R., Ponsford, J. L., Hicks, A. J., Hopwood, M., Renison, B., & Feeney, T. J. (2021). Positive behaviour support for challenging behaviour after acquired brain injury: An introduction to PBS + PLUS and three case studies. *Neuropsychological Rehabilitation*, 31(1), 57–91. <https://doi.org/10.1080/09602011.2019.1656647>
- Hassiotis, A., Poppe, M., Strydom, A., Vickerstaff, V., Hall, I., Crabtree, J., Omar, R., King, M., Hunter, R., Bosco, A., Biswas, A., Ratti, V., Blickwedel, J., Cooper, V., Howie, W., & Crawford, M. (2018). Positive behaviour support training for staff for treating challenging behaviour in people with intellectual disabilities: A cluster RCT. *Health Technology Assessment*, 22(15), 15. <https://doi.org/10.3310/hta22150>
- Hayward, B. A. (2021). Where is applied behaviour analysis (ABA) in the National Disability Insurance Scheme? Commentary on “Delivering behaviour support to children and adolescents with autism via telepractice”: A narrative review (Johnsson & Bulkeley, 2021). *Research and Practice in Intellectual and Developmental Disabilities*. Advance online publication. <https://doi.org/10.1080/23297018.2021.1937292>
- Hayward, B. A., McKay-Brown, L., & Poed, S. (2019a). Positive behaviour support in Australian disability policy, and its future with the National Disability Insurance Scheme (NDIS). *Research and Practice in Intellectual and Developmental Disabilities*, 6(1), 14–23. <https://doi.org/10.1080/23297018.2018.1498299>
- Hayward, B. A., McKay-Brown, L., & Poed, S. (2019b). Reply to Higgins’ (2019) commentary on “Positive behaviour support in Australian disability policy, and its future with the National Disability Insurance Scheme (NDIS) (Hayward, McKay-Brown, & Poed, 2019). *Research and Practice in Intellectual and Developmental Disabilities*, 6(1), 36–38. <https://doi.org/10.1080/23297018.2019.1595098>
- Hayward, B. A., Poed, S., & McKay-Brown, L. (2017). Analysing text in positive behavioural support (PBS) policy: A pilot study. *International Journal of Positive Behavioural Support*, 7(2), 37–48.
- Hayward, B. A., Poed, S., & McKay-Brown, L. (2018). Improving the adoption of PBS and ABA using diffusion of innovations theory. *Tizard Learning Disability Review*, 23(4), 178–186. <https://doi.org/10.1108/TLDR-02-2018-0005>
- Higgins, A. (2019). Commentary on “Positive behaviour support in Australian disability policy, and its future with the National Disability Insurance Scheme (NDIS) (Hayward, McKay-Brown, & Poed, 2019). *Research and Practice in Intellectual and Developmental Disabilities*, 6(1), 24–35. <https://doi.org/10.1080/23297018.2019.1581581>
- Himelboim, I., Smith, M. A., Rainie, L., Shneiderman, B., & Espina, C. (2017). Classifying Twitter topic-networks using social network analysis. *Social Media + Society*, 3(1), 1–13. <https://doi.org/10.1177/2056305117691545>
- Horner, R. H. (2000). Positive behavior supports. *Focus on Autism and Other Developmental Disabilities*, 15(2), 97–105. <https://doi.org/10.1177/108835760001500205>
- Horner, R. H., & Sugai, G. (2018). Future directions for positive behavior support: A commentary. *Journal of Positive Behavior Interventions*, 20(1), 19–22. <https://doi.org/10.1177/1098300717733977>
- Jesiek, B. K., Mazzurco, A., Buswell, N. T., & Thompson, J. D. (2018). Boundary spanning and engineering: A qualitative systematic review. *Journal of Engineering Education*, 107(3), 380–413. <https://doi.org/10.1002/jee.20219>
- Johnson, J. D. (2012). The role of human agents in facilitating clinical and translational science. *Clinical and Translational Science*, 5(5), 356–361. <https://doi.org/10.1111/j.1752-8062.2011.00379.x>
- Klov Dahl, A. S. (2005). Social network research and human subject protection: Towards more effective infectious disease control. *Social Networks*, 27(2), 119–137. <https://doi.org/10.1016/j.socnet.2005.01.006>
- Kowald, M., & Axhausen, K. W. (2012). Focusing on connected personal leisure networks: Selected results from a snowball sample. *Environment and Planning A*, 44(5), 1085–1100. <https://doi.org/10.1068/a43458>
- Laumann, E. O., Marsden, P. V., & Prensky, D. (1983). The boundary specification problem in network analysis. In R. Burt & M. Minor (Eds.), *Applied network analysis* (pp. 18–34). Sage.
- Lewis, J. M. (2006). Being around and knowing the players: Networks of influence in health policy. *Social Science & Medicine*, 62(9), 2125–2136. <https://doi.org/10.1016/j.socscimed.2005.10.004>
- Long, J. C., Cunningham, F. C., & Braithwaite, J. (2013). Bridgers, brokers and boundary spanners in collaborative networks: A systematic review. *BMC Health Services Research*, 13(1), 158. <https://doi.org/10.1186/1472-6963-13-158>
- Malbon, E., Alexander, D., Carey, G., Reeders, D., Green, C., Dickonson, H., & Kavanagh, A. (2017). Adapting to a marketised system: Network analysis of a personalisation scheme in early implementation. *Health and Social Care in the Community*, 27(1), 191–198. <https://doi.org/10.1111/hsc.12639>
- McGill, P., Papachristoforou, E., & Cooper, V. (2006). Support for family carers of children and young people with developmental disabilities and challenging behaviour. *Child: Care, Health and Development*, 32(2), 159–165. <https://doi.org/10.1111/j.1365-2214.2006.00600.x>
- McGill, P., Vanono, L., Clover, W., Smyth, E., Cooper, V., Hopkins, L., Barratt, N., Joyce, C., Henderson, K., Sekasi, S., Davis, S., & Deveau, R. (2018). Reducing challenging behaviour of adults with intellectual disabilities in supported accommodation: A cluster randomized controlled trial of setting-wide positive behaviour support. *Research in Developmental Disabilities*, 81, 143–154. <https://doi.org/10.1016/j.ridd.2018.04.020>
- NDIS Quality and Safeguards Commission. (2019). *Positive behaviour support capability framework: For NDIS providers and behaviour support practitioners*. <https://www.ndiscommission.gov.au/pbscapabilityframework>
- Nott, M. T., Barden, H. L. H., Chapparo, C., & Ranka, J. L. (2019). Evidence based practice and knowledge translation:

- A survey of Australian occupational therapy practice with clients experiencing neurocognitive impairments. *Australian Occupational Therapy Journal*, 67(1), 74–82. <https://doi.org/10.1111/1440-1630.12625>
- O'Connor, C. M. C., Mioshi, E., Kaizik, C., Fisher, A., Hornberger, M., & Piguet, O. (2019). Positive behaviour support in frontotemporal dementia: A pilot study. *Neuropsychological Rehabilitation*, 31(4), 507–530. <https://doi.org/10.1080/09602011.2019.1707099>
- O'Dwyer, C., McVilly, K. R., & Webber, L. (2017). The impact of positive behavioural support training on staff and the people they support. *International Journal of Positive Behavioural Support*, 7(2), 13–23.
- Parsons, B. M. (2018). Local autism policy networks: Expertise and intermediary organizations. *Educational Policy*, 32(6), 823–854. <https://doi.org/10.1177/0895904816673743>
- Perez, M., Carlson, G., Ziviani, J., & Cuskelly, M. (2012). Contribution of occupational therapists in positive behaviour support. *Australian Occupational Therapy Journal*, 59(6), 428–436. <https://doi.org/10.1111/j.1440-1630.2012.01036.x>
- Rillotta, F., & Alexander, J. (2019). Roles and responsibilities of developmental educators: Disability professionals supporting Australians with disability. *International Journal of Disability, Development and Education*. Advance online publication. <https://doi.org/10.1080/1034912X.2020.1719987>
- Rogers, E. M. (2003). *Diffusion of innovations* (5th ed.). Free Press.
- Rotholz, D. A. (2009). Creating the future: Beyond our inheritance of past. *Intellectual and Developmental Disabilities*, 47(2), 125–134. <https://doi.org/10.1352/1934-9556-47.2.125>
- Sabot, K., Wickremasinghe, D., Blanchet, K., Avan, B., & Schellenberg, J. (2017). Use of social network analysis methods to study professional advice and performance among healthcare providers: A systematic review. *Systematic Reviews*, 6(1), 208. <https://doi.org/10.1186/s13643-017-0597-1>
- Soo, C., Chen, S., & Edwards, M. G. (2017). A knowledge-based approach to public value management: A case study of change implementation in disability services in Western Australia. *Australian Journal of Public Administration*, 77(2), 187–202. <https://doi.org/10.1111/1467-8500.12279>
- Tasselli, S. (2014). Social networks of professionals in health care organizations: A review. *Medical Care Research and Review*, 71(6), 619–660. <https://doi.org/10.1177/1077558714557079>
- Trepte, S., & Scherer, H. (2010). Opinion leaders – do they know more than others about their area of interest? *Communications*, 35(2), 119–140. <https://doi.org/10.1515/COMM.2010.007>
- Vassos, M., & Carroll, M. F. (2016). Assessing attitudes toward evidence-based practices of workers supporting people with disabilities: A validation of the evidence-based practice attitudes scale. *American Journal on Intellectual and Developmental Disabilities*, 121(4), 364–378. <https://doi.org/10.1352/1944-7558-121.4.364>
- Vassos, M., Nankervis, K., & Chan, J. (2019). *Defining the work of a disability support worker in Australia: A review of the work role*. <https://www.researchgate.net/project/Defining-the-work-of-a-disability-support-worker-DSW-A-systematic-review-of-the-DSW-work-role>
- Wasserman, S., & Faust, K. (1994). *Social network analysis. Methods and applications*. Cambridge University Press.
- Webber, L., Makor, K., Condello, C., & Hancox, K. (2017). Providing positive behaviour support to improve a client's quality of life. *Learning Disability Practice*, 20(4), 36–41. <https://doi.org/10.7748/ldp.2017.e1826>
- Williams, P. (2011). The life and times of the boundary spanner. *Journal of Integrated Care*, 19(3), 26–33. <https://doi.org/10.1108/14769011111148140>
- Williamson, K. (2018). Populations and samples. In K. Williamson & G. Johanson (Eds.), *Research methods. Information, systems, and contexts* (pp. 359–377). Chandos.

Appendices

Appendix 1. Adapted appraisal tool for social network analyses [from Sabot et al. (2017)]

Topic	Item No.	Response
Theoretical approach		
Scientific background and rationale for the study	1	Addresses existing research recommendations and current Australian disability context
Research aims/objectives/research questions/hypotheses	2	Three aims listed
Theory underpinning the study design discussed	3	Description of population to be studied based on included literature review; SNA and use of opinion leaders and boundary spanners outlined
Methods		
Research design/methodology rigorous and focused	4	Hidden populations; non-probability sample
Methods and data collected appropriate for research	5	Survey methodology used in previous SNA studies; use of a seed sample; online to facilitate ease of participation
Explanation given for how the study size was determined	6	Relational approach to establish social boundary
Outcomes, sources and data collection methods described	7	All described in detail; study participants described
Rigorous selection of study participants	8	Reputational snowball sampling
Efforts made to reduce bias during data collection	9	Missing and incorrect data management explained; definitions of boundary spanners and opinion leaders applied
Study duration and response rate/follow up	10	Outlined in the text and Figure 1
Results		
Findings clearly presented	11	Survey participants, network graph, sociometric measures, demographics, and network positions presented
Analyses conducted to account of possible sources of bias	12	Not relevant to this study
Results valid/precise/believable	13	Data for boundary spanners and opinion leaders matched to criteria
Social network analysis		
Network boundaries clearly defined and appropriate	14	Social boundary established through a reputational snowball procedure
Appropriate SNA metrics selected, defined, and calculated	15	Betweenness centrality, indegree centrality, outdegree centrality
Response rates given for whole network studies	16	Included
Definition of tie relationships, direction and strength	17	Relationships explained through description of network positions (Table 1); outdegree included; strength not relevant to this study
Discussion		
Link between data, interpretation and conclusions	18	Results matched to criteria for network positions and Australian context
Alternative explanations explored and discounted	19	Criteria not met explained in context of immature system; options for types of opinion leaders described
Other studies with similar/different results discussed	20	Yes, including disability context, PBS, and methodological comparisons
Implications of the research discussed	21	Yes, for both network positions and for the application of PBS in disability services; further research suggestions described
Discussion of study limitations	22	Included
Other		
Approved by an ethics committee	23	Yes
Funding source for the study clearly stated	24	Yes, but not applicable.

Note: SNA = social network analysis.

Appendix 2. Adapted reporting guidelines for survey research [from Bennett et al. (2011)]

Topic	Item No.	Response
Title and Abstract		
Design of the study stated in the title and/or abstract	1	Yes, in title
Introduction		
Explanation of why the research is necessary	2	Yes, for PBS, disability services, and Australia
Purpose or aim of the paper explained	3	Aims explicitly listed
Methods		
<i>Research Tool</i>		
Questionnaire described	4	Items listed in Appendix 1 and Table 2
Psychometric properties of tools presented	5	Not relevant for this study

(Continued)

Continued.

Topic	Item No.	Response
References to original and new tools provided	6	Not relevant for this study
Description of the scoring procedures provided	7	Application of criteria for network positions outlined in methods section and Table 1
<i>Sample Selection</i>		
Description of the survey population and the sample frame used to identify this population	8	Non-probability sample; hidden population
Description of how representative the sample is of the underlying population	9	Population is specified in terms of a person's presence in the network, their attributes, and the content of their relationships
Sample size calculation or rationale for the sample size presented	10	Sample size is not relevant for this study; reputational snowball sampling used to establish social boundary
<i>Survey Administration</i>		
Mode of administration stated	11	Seed sample face-to-face; snowball online
The type of contact and how many attempts were made to contact subjects	12	Face-to-face for seed sample at one time; email for snowball with one reminder
Report whether incentives were provided	13	Incentives used and outlined in Context section
Description of who approached potential participants	14	One approach via third party for seed sample; two contacts by first author for snowball
<i>Analysis</i>		
Method of data analysis described	15	Using Qualtrics® online survey platform; spreadsheets for analysis
Methods for analysis of nonresponse error	16	Described in Figure 1
Method for calculating response rate provided	17	Percentages
Definitions provided for complete versus partial completions	18	Included in data extraction and analysis section of paper
Methods for handling missing data provided	19	Included in data extraction and analysis section of paper
Results		
Response rate reported	20	Yes
All respondents accounted for	21	Yes
Information given on how non-respondents differ from respondents	22	Outlined in Figure 1
Results clearly presented	23	Survey participation and demographics reported
Results address the objective(s)	24	Aim no. 2 met
Discussion		
Results summarised with reference to the study objectives	25	All three aims are addressed through application of the results in the discussion
Strengths and limitations of the study stated	26	Included in dedicated section
Explicit discussion of the generalisability of the results	27	Included in both the introduction and discussion
Ethical Quality Indicators		
Study funding reported	28	Yes, although this is not applicable in this study
Research Ethics Board review reported	29	Yes
Reporting of subject consent procedures	30	Yes

Appendix 3. Self-reported characteristics of boundary spanners, opinion leaders, and those in both network positions

Characteristic	Boundary Spanner <i>n</i> = 9	Opinion Leader <i>n</i> = 4	Both <i>n</i> = 6
Job title			
Other ^a	2	2	3
Clinical ^b	3	1	2
Education	1	0	0
Manager	1	0	1
Academic	2	1	0
Job type			
Other	0	0	1
Government	3	0	2
Support organisation	4	0	0
Private	0	3	3
University	2	1	0
Education			
Postgraduate degree	5	3	5
PhD or doctorate	3	0	1
Undergraduate	1	1	0
Years working in disability sector			
1–5	2	0	0
6–10	1	0	0
11–20	2	3	2
>20	4	1	4
Number of professional memberships			

(Continued)

Continued.

Characteristic	Boundary Spanner <i>n</i> = 9	Opinion Leader <i>n</i> = 4	Both <i>n</i> = 6
None	3	3	1
One	3	1	2
Two	2	0	1
Three	1	0	2
Scope of work			
International only	1	0	1
State-wide ^c	2	2	2
Multi-state ^d	1	1	3
Local ^e	3	0	0
National ^f	2	1	0
Years working in current role			
<1	1	0	2
1–5	4	2	1
6–10	1	1	1
11–20	2	1	2
Primary location of work			
International	0	0	1
Melbourne CBD	1	0	2
Metropolitan Melbourne	6	2	2
Other state	2	0	1
Regional Victoria	0	2	0
Work type			
Education and training	2	1	1
Information and advice	1	0	1
Assessment & planning	3	1	3
Management	2	1	1
Research	1	1	0
Primary disability group focus			
No direct work	4	0	2
Autism spectrum disorder	3	1	1
Intellectual disability	2	2	3
Acquired brain injury	0	1	0
Primary disability age focus			
No direct work	4	0	2
0–5	1	0	0
6–18	1	0	1
19–64	3	4	3

^aThis was an answer option in the survey. ^bIncludes medical practitioner (*n* = 2), nurse (*n* = 2), occupational therapist (*n* = 3), social worker (*n* = 4) and speech pathologist (*n* = 5). ^cAll of [BLIND] only. ^d[BLIND] plus one or more other Australian state or territory. ^eA defined geographical area of [BLIND]. ^fAll Australian states & territories.

Appendix 4. Raw data for sources of information.

Respondent	Websites	Social	Journals	Books	Reports	Conference	Training	Networks	Total	Mean	>network position mean
48	3	1	3	3	2	3	2	2	19	2.375	Y
40	3	2	3	3	3	2	2	3	21	2.625	Y
30	2	0	1	1	1	1	1	1	8	1	N
06	1	3	4	1	0	1	2	2	14	1.75	N
53	3	3	2	2	1	1	1	1	14	1.75	N
09	2	3	3	3	1	1	1	1	15	1.875	Y
36	3	4	3	2	1	1	1	2	17	2.125	Y
05	2	2	3	1	1	1	1	1	12	1.5	N
41	3	0	3	3	3	3	3	3	21	2.625	Y
66	3	4	3	3	2	2	2	3	22	2.75	Y
28	3	0	3	3	3	2	1	3	18	2.25	Y
04	2	0	2	2	2	1	1	2	12	1.5	N
17	0	0	1	1	1	1	1	1	6	0.75	N
26	3	4	1	2	1	1	1	3	16	2	N
21	3	4	3	3	2	1	1	2	19	2.375	Y
01	4	4	3	2	3	1	1	2	20	2.5	Y
43	3	0	3	3	3	3	3	3	21	2.625	Y
11	2	2	2	2	2	2	2	2	16	2	N
46	3	3	2	2	2	1	1	2	16	2	N

Note: Means for network positions: opinion leaders = 2; boundary spanners = 3; combined = 3.